VF-E02 Colorfastness to Crocking

Test Method for Colorfastness to Crocking	Test method # VF- E02 REF # : TM-R- 011
Date Issued : August 12, 2024 Revision # : 01	Reason for revision: Change solvent test to artificial perspiration test for paint/print

PURPOSE AND SCOPE: To measure the tendency of dyestuffs and other colorants to migrate from a test material, as a result of rubbing.

REFERENCE: AATCC 8

EQUIPMENT AND MATERIAL:

1. Crockmeter, meeting the requirements of AATCC Test Method 8.
2. Crockmeter test cloth, meeting the requirements of AATCC Test Method 8
3. Immersion dish.
4. Distilled water
5. Artificial perspiration solution (PH 4.3) for ink
6. AATCC Chromatic Transference Scale
7. Fabric, Leather, packaging, paint, print and synthetic leather.

SAMPLE PREPARATION:

Condition the sample for at least 24 hours at the standard laboratory environment @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A. Test Material

Die or hand cut two test samples to fit be entire platform of the crockmeter, one sample will be used for dry crocking and the other for wet crocking.

B. Crockmeter Test Cloth

Die or hand cut a piece of crockmeter test cloth approximately 5cm*5cm.

REPETITIONS: Test one (1) dry sample and one (1) wet sample per material.

TEST PROCEDURE:

Test should be conducted at standard laboratory conditions @ 23.0±2.0˚C temperature and 55 ±10 % Relative Humidity.

A. Dry Test:

1. Place the test material on the base of the crockmeter, and secure by positioning the holder over the material and clamping.
2. Mount the crockmeter test cloth on the end of the 1.5cm diameter finger of the crockmeter and clamp in place.
3. Lower the covered finger onto the test specimen (the crockmeter head should weigh 900grams).

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4. i) Manual type crockmeter: Reference GT-7034 B (Gotech) or equivalent crockmeter meeting the requirement of AATCC Test method 8

   - Crank the crockmeter handles for 10 complete (back and forth) passes over the surface of the test material, (The distance of each pass should be 10.4+/-0.3cm, the passes should be made at a rate of approximately 1 pass /second.)

   ii) Motorized type crockmeter: Reference GT-7034 A (Gotech) or equivalent crockmeter meeting the requirement of AATCC Test method 8

   - Set the timer to 10 cycles. Initiate testing.

5. At the end of 10 cycles, remove the crockmeter test cloth from the finger and place on a white piece of paper.

Note: Before evaluating the test sample, remove any extraneous fibrous material by pressing lightly on the crock circle with the sticky side of a piece of cellophane tape.

6. Evaluate the colorfastness to dry crocking by comparing the darkest portion of the test cloth to the AATCC Chromatic Transference Scale. The evaluation must be performed under in the light box, set for 'Daylight' (D65).

Note: Chromatic Transference Scales must be replaced at least every 12 months.

B. Wet Test

1. Place the test material on the base of the crockmeter, and secure by positioning the holder over the material and clamping.

Note: Do not use the same area of the fabric which was used for the dry test

2. Thoroughly wet the crockmeter test fabric by dipping in distilled water, allow any excess water to drip from the test cloth, position on the finger of the crockmeter and clamp in place.
3. Immediately lower the covered finger onto the test specimen.
4. Crank the crockmeter handles for 10 complete (back and forth) passes over the surface of the test material, (The distance of each pass should be 10.4+/-0. 3cm.The passes should be made at a rate of approximately 1 pass /second).
5. Remove the crockmeter cloth and allow to fully dry at 23+/-2°C.

Note: Before evaluating the test sample, remove any extraneous fibrous material by pressing lightly on the crock circle with the sticky side of a piece of cellophane tape.

6. Evaluate the colorfastness to wet crocking as described in **STEP 6** of the dry test.

C. Artificial Perspiration Test (for paint/print contact with skin)

This test is performed in an identical manner as the **"Wet-Test",** except the crockmeter test fabric is dipped in the designated artificial perspiration solution (PH4.3), instead of distilled water.

REPORT: Both the dry and wet crocking result to the nearest 0.5, according to the AATCC Chromatic Transference Scale.